A1: Part 2 Loading into Data Viz | Michel Ward

Oct 24, 2026

Part A: Dashboard Screenshots

The interactive dashboard is published on Tableau Public: Client #128 [Portfolio Dashboard](). Each screenshot below shows a different filter or highlight state, and each one supports a finding in Part B.

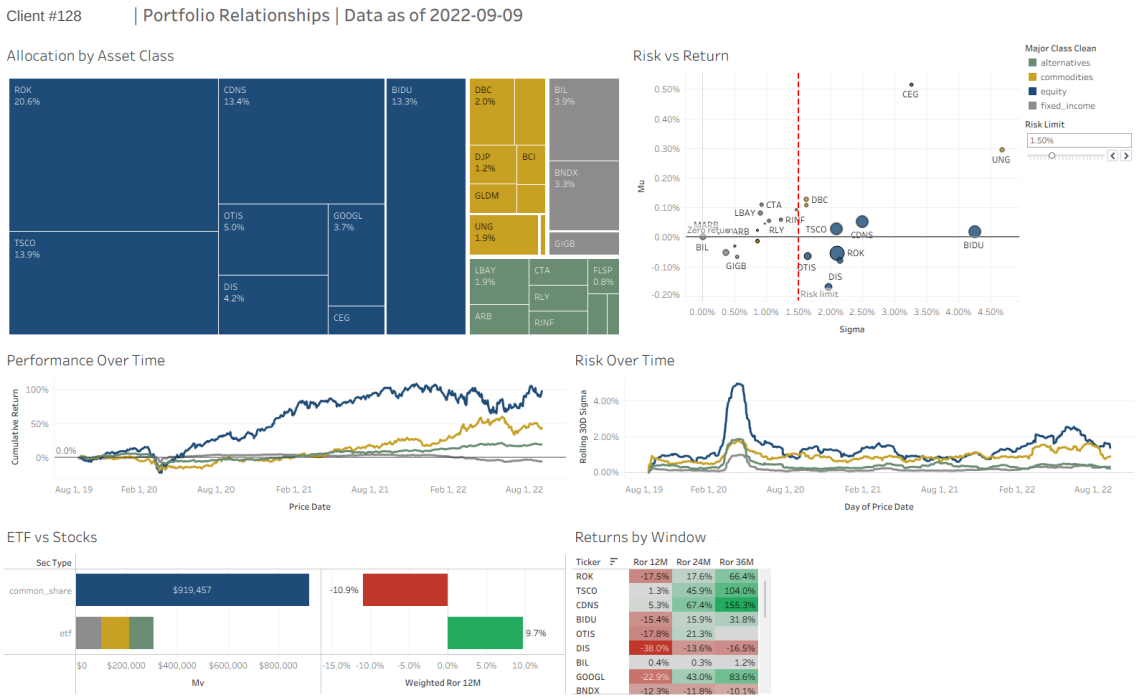


Figure 1: Full portfolio view with the Risk Limit parameter moved to 1.5%. Every stock now sits to the right of the red line.

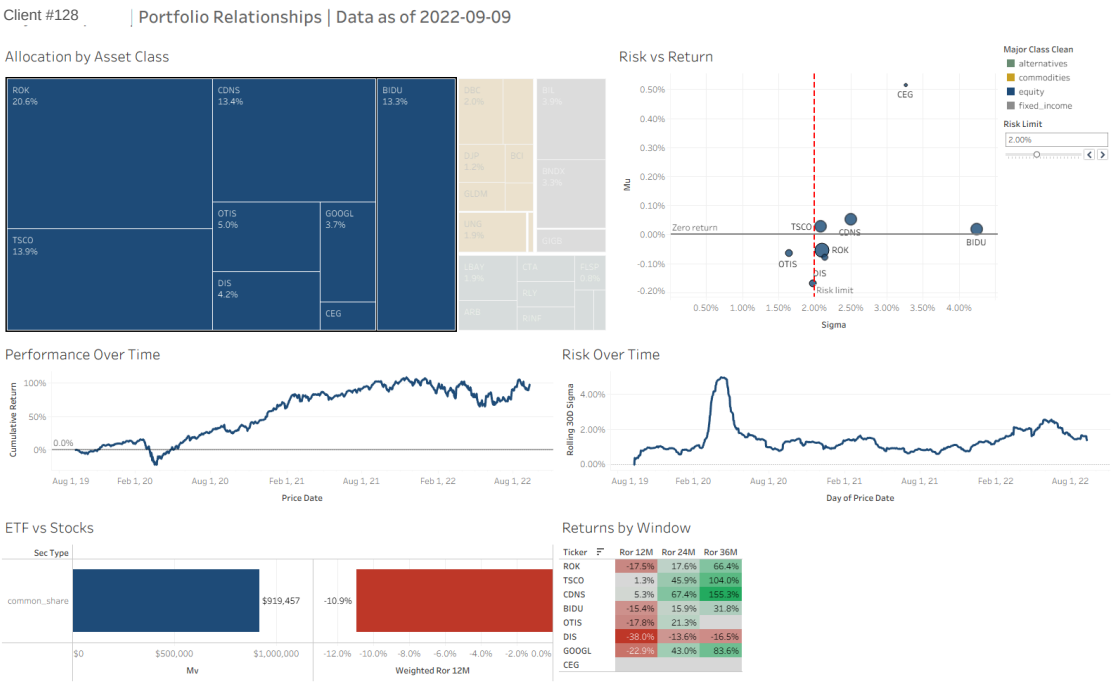


Figure 2: Equity selected in the treemap. The filter action updates every other chart to show only her stocks.

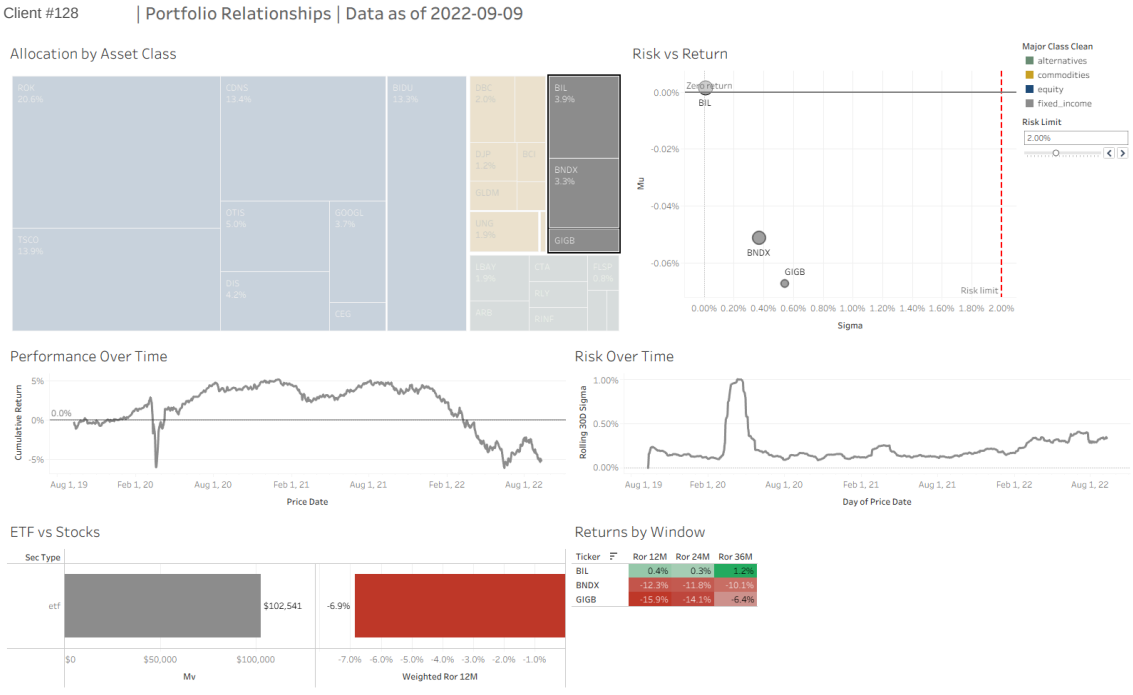


Figure 3: Fixed income selected in the treemap. Only BIL held its value; BNDX and GIGB lost money in every window.

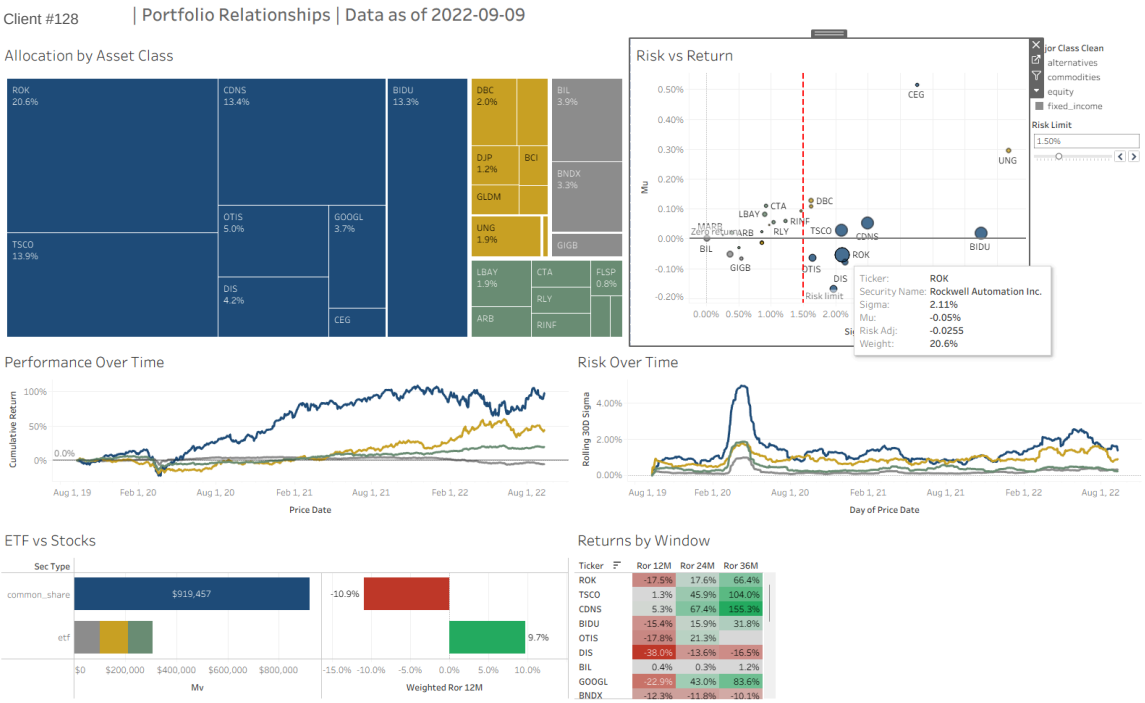


Figure 4: Hovering ROK highlights it across the dashboard. The tooltip shows her largest position: 20.6% weight, sigma 2.11%, negative mu and a risk adjusted return of −0.0255.

Part B: Report for Management

1. Introduction and Approach

Who is my client? My client is Client #128 customer #128 from the Hult Dubai campus. She has two accounts, a main account opened in December 2018 and a sub account opened in April 2020. Together they are worth $1,225,776 across 27 securities. All the data is as of 9 September 2022.

Where the data comes from. In Part 1, I used SQL to calculate her returns, risk and risk adjusted returns. For Part 2, I wanted the dashboard to show exactly the same numbers, so I exported both datasets from my Part 1 view, invest.michel_ward, and added the market value weights from Question 1. This is the ETL idea we saw in class: if I only extract prices, I get findings. When I transform them into returns and risk before loading them into Tableau, I get insights.

How I designed the dashboard. I tried to follow the data viz rules from class: keep it simple, be consistent and highlight what matters. Each asset class has the same color in every chart: navy for equity, gold for commodities, green for alternatives and gray for fixed income. I used red and green only for returns. I also added three interactions: clicking the treemap filters every chart, hovering a ticker highlights it everywhere, and a Risk Limit slider moves a red line on the risk vs return chart.

2. Findings

Figure 1: Most of her money sits in a few risky stocks. The first thing I noticed in the treemap is how much space the navy blocks take. Stocks hold $919,457, about 75% of her portfolio, and only four of them, ROK (20.6%), TSCO (13.9%), CDNS (13.4%) and BIDU (13.3%), hold 61% of her money. When I

moved the Risk Limit to 1.5% daily sigma, every single stock ended up on the right side of the line. In simple terms, three quarters of her wealth is above that risk level.

Her ETFs did better than her stocks. Over the last 12 months, her stocks lost 10.9%, while her ETFs ($306K) gained 9.7%. This is the same story as Part 1, where equity was the reason the whole portfolio lost 5.9%.

Time changes the picture. Equity's cumulative return reached about +108% in late 2021, dropped to about +65% by June 2022, and ended near +97%. Its rolling 30 day sigma jumped to about 5% during COVID in March 2020, and to about 2.5% again in mid 2022. Meanwhile, commodities climbed to about +60% in 2022, and fixed income was always the calmest line.

Figure 2: The stocks that made her money are the same ones that lost it. When I selected equity, the heat table showed great 36 month returns: CDNS +155%, TSCO +104%, GOOGL +84% and ROK +66%. But the 12 month column is mostly red: GOOGL −22.9%, OTIS −17.8%, ROK −17.5% and BIDU −15.4%. DIS is red in every window, with −38.0% over 12 months. Figure 4 shows ROK on its own: the biggest bubble below zero return, with a sigma of 2.11% and a risk adjusted return of −0.0255. This is the same risk without reward I found in Part 1.

Figure 3: Her bonds did not protect her. With fixed income selected, only BIL held its value (+0.4% over 12 months). BNDX lost 12.3% and GIGB lost 15.9%. The performance line went up about 5% until late 2021, then dropped below zero in 2022 when interest rates started rising. Weighted by market value, her fixed income lost 6.9% over 12 months. Her "safe" layer was not safe this year.

She owns the same gold three times. In the scatter, GLDM, IAUM and SGOL sit on exactly the same point, which confirms what I found in Part 1.

3. Business Insights

A few stocks are both her engine and her biggest risk. Four stocks created about 90% of her 36 month gain, and the same four caused most of her 12 month loss. So even though she holds 27 securities, her portfolio does not really behave like a diversified one. It behaves like a bet on a few companies.

I don't think she chose this allocation. CDNS and TSCO more than doubled, so their weight grew by itself. My reading is that the 75% in stocks is the result of growth that was never rebalanced, not a deliberate decision. In wealth management, where we think in 10 year horizons, that is a risk management problem, not just a good or bad year.

Her bonds were protecting her against the wrong risk. Bonds are supposed to help when stocks fall. In 2022, both fell together because rising rates hurt both. BNDX and GIGB hold longer duration bonds, so they were exposed to exactly what happened. BIL, which is very short term, was the only one that worked. For me, the lesson is that a safe asset is only safe against the risk it was chosen for.

Diversification worked, it was just too small. As we discussed in class, commodities have weak correlation with other asset classes, and her commodity ETFs went up exactly when her stocks went down. The problem is size: at about 9% of the portfolio, they could not change the result.

Risk comes in waves. The rolling sigma chart shows long calm periods broken by sharp spikes in 2020 and 2022. If Client only looked at her average risk, she would underestimate how fast her portfolio can move. Seeing risk over time tells us when protection is needed, not only how much.

The data cannot tell me what she wants. In class, we learned that the first client meeting covers current assets, future goals and risk aversion. My analysis covers the first part very well, but not the other two. Before changing anything, I need to hear from her.

4. Recommendations

Every recommendation below comes from something I saw in the dashboard, and I point to the chart that shows it.

1. **Cut ROK from 20.6% to about 10% and move $122,578 into VTIP.** ROK is the largest bubble below zero return (Figure 4). In Part 1, I simulated this exact move, and her portfolio sigma dropped from 1.40% to 1.25%, about 10.5% less risk, while her average return went up.
2. **Sell DIS ($51,370) and use part of the money to double LBAY.** DIS is red in every window of the heat table. LBAY had her best full year risk adjusted return (0.091), with less risk than any of her stocks.
3. **Replace BNDX and GIGB ($54,519) with short duration bonds like VTIP or SHY.** Figure 3 shows the longer bonds failed in 2022, while short term BIL held its value. Her protection should match the risk she is actually facing.
4. **Keep only one gold ETF.** GLDM, IAUM and SGOL are the same point in the scatter. Moving about $34,850 into the one with the lowest fees keeps the same exposure with less complexity.
5. **Take some profits in CDNS and TSCO, step by step.** Trimming each to 10% frees about $90,000 without selling her winners completely. I would do it over two quarters. Since she is based in Dubai, I would confirm her tax situation first.
6. **Put those profits into a broad market ETF like SPY.** Her ETFs beat her stocks (+9.7% vs −10.9%). A broad fund keeps her exposure to growth, but spread across hundreds of companies instead of four.
7. **Use the dashboard every quarter, with a simple risk policy.** I would suggest two rules: no single stock above 10%, and a review every time a holding crosses the Risk Limit line. Because the limit is a parameter, it can be adjusted to her risk tolerance.
8. **Talk to** Client **before doing any of this.** Following the first meeting framework from class, I would confirm her goals, time horizon and risk aversion. If she wants growth over 10 years, recommendations 5 and 6 can be smaller. If she wants to protect what she has built, I would do them in full.

What I expect: after these changes, no stock would be above 10%, single stocks would go from about 75% to about 55% of her money, and her bonds would be ready for rising rates. Based on Part 1, her risk should drop meaningfully without losing her long term growth.

5. Conclusion

Client **has done very well, but her gains depend on too few stocks.** For every $100 she had invested in September 2019, she had about $168 in September 2022. What this dashboard showed me is how to

protect that: fewer big single bets, bonds that fit today's risk, and a regular check on how risk changes over time.

A note on the method. The two time charts use an equal weighted average of the holdings in each asset class. That is why equity shows about +97% over 36 months there, compared with +80.5% in Part 1, where I weighted by market value. The numbers are different, but both tell the same story.

Appendix

Data sources. Two views built on my Part 1 view invest.michel_ward, exported from MySQL Workbench as CSV files and loaded into Tableau Public with a relationship on ticker:

File	Built from	Rows	Used for
summary.csv	invest.michel_ward_summary	27, one per security	Treemap, risk vs return, ETF vs stocks, returns by window
daily.csv	invest.michel_ward_dash	One per security per trading day	Performance and risk over time

Data cleaning in Tableau. The database spells fixed income in two ways, so I created a calculated field to merge them. Missing minor asset classes take the major class name, and the typo "aggriculture" was corrected.

Dashboard link: Client #128 Portfolio Dashboard on Tableau Public

AI use. I used Claude to help write the SQL views, plan the dashboard and structure this report. Claude conversation used for this assignment: https://claude.ai/artifact/1xasPcvfcph6yUQtGmaezb#1569deb2-3694.mw1gfwryt67.8962